

net of Things (IoT) and other emerging issues will be taken up.

## Electrical communication

“Man is the only animal that laughs and weeps; for he is the only animal that is struck with the difference between what things are, and what they ought to be,” said William Hazlitt, the English comic writer, in 1819.

When a baby is born, what it does at first is a form of communication, that is, weeping. This communication is natural voice communication. Natural voice (face-to-face) communication also involves video communication—as, when we speak, we see too. Besides, when we speak and see, we transmit data. Hence, this type of communication can also be called natural multimedia (voice + video + data/text) communication.

Any communication system must have three basic units: transmitter, receiver and transmission line. In natural voice communication, when we speak, our mouth acts as the transmitter and when we hear, our ear acts as the receiver. In between these two, air acts as the transmission line.

Natural voice communication has a few unique characteristics, as given below.

- It is instantaneous, with virtually zero delay between the transmitter and receiver. Thus, it is also real-time communication.
- This type of communication can be best described as “When I hear you, I feel you and I see you,” and so “When I speak with you, you feel me and you see me.” Thus, natural voice communication is more than just hearing (voice). It is seeing (video) and feeling (perception), too.
- It supports man-mobility. As we move, we can speak, see and feel, too.
- It is interactive.
- It is one where identity of the transmitter and receiver never differs from place to place. For example, your name does not change as you move from one place to another.
- It is fully wireless.

Natural voice communication is not without its limitations. It is essentially a short-haul form of communication. Its coverage area is small. Using this mode, communication beyond a few



A Morse key, circa 1900 (Credit: upload.wikimedia.org)

metres is not possible.

Many scientific and technological developments are aimed to overcome such challenges.

## The telegraph: early long-distance text communication

Ancient civilisations of China, Greece and Egypt, for the purpose of long-distance communication, used drum-beats or smoke signals for exchange of information. As an improvement to such primitive systems, the semaphore was developed in early 1700s. In the semaphore system, a series of hilltops was used as movable arms to signal letters and numbers. Two telescopes were used to see the other stations. Later, improvements were made only in terms of increasing the coverage area of communication.

These ancient systems were susceptible to failures of line-of-sight between communicating stations, low visibility and bad weather, among others.

Subsequently, various other forms of communication were developed for long-distance communication to exchange information, ideas, thoughts, etc. A few examples are letters, books and newspapers.

Over the years, many systems of exchanges have been developed—including postal systems and newspapers/books delivery systems—basically as man-managed systems. All these developments attempted to achieve long-distance communication but could not attain any unique characteristic of natural voice communication. However, in 1790s came the opportunity to attain the desired goal with the invention of the first electrical communication, by the name of telegraph. On March 2, 1791, the first telegraphic message was sent.



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